

Iris Output Data Specification



1 Introduction

This document describes the packet format that is used to output lidar data over Ethernet from the Iris sensor.

This data format is largely compatible with the spirit of Draft ISO/DIS 23150 (Road vehicles – Data communication between sensors and data fusion unit for automated driving functions -- Logical interface), but differs in specific details. The reasons for these differences are twofold: 1) to reduce bandwidth requirements by removing duplicate data and 2) to reduce bandwidth requirements by removing fields that the Iris sensor does not calculate. In practice, this means that the azimuth and elevation angles are specified per ray, rather than per return, and that angle error, range error, and reflectance error are not reported.

1.1 Version

Both this document and the packet format rely on semantic versioning [<https://semver.org/>]. The version number of this document and of the packet format match.

This document describes version 1.3.0 of the packet format.

1.2 Nomenclature

1.2.1 Fixed Point Values

Fixed point values are represented with an unambiguous notation $[U]Q_m.n$.

- If specified before the Q , U indicates that the value should be interpreted as unsigned. Otherwise, the value is signed.
- m specifies the number of integer bits in the value
- n specifies the number of fractional bits
- The total number of bits in the value is $m+n$
- A signed value must have $m > 0$

Examples:

- $Q8.0$ is equivalent to `stdint.h` type `int8_t`
- $UQ16.0$ is equivalent to `stdint.h` type `uint16_t`
- $Q5.3$ is an 8 bit (5+3) signed type, representing the range -16.0 to 15.875 with a resolution of 0.125
- $UQ0.10$ is a 10 bit unsigned type, representing the range 0 to 0.99902 with a resolution of 2^{-10}

2 Data Specification

The Ethernet data packet has the structure:

Data Field
UDP Packet Header
Lidar Packet Header
Ray Header 1
Return Data 1
Return Data 2
Return Data 3
Ray Header 2
Return Data 1
Return Data 2
Ray Header 3
Return Data 1
Return Data 2
Return Data 3
Return Data 4
...
Data Protection Footer

Note that the packet may contain zero or more rays (specified in the lidar packet header), and that each ray may contain zero or more returns (specified in the ray header).

Packets will be sent, and a new packet started when any of the following occur:

- The packet reaches the Ethernet MTU
- A one second boundary is reached
- A new scan frame begins
- The data qualifier field changes
- The scan identifier changes

All data from a previous second, frame, or data qualifier will be transmitted before any data from the new second, frame, data qualifier, or scan identifier.

If not otherwise specified, all data fields are represented in 64 bit little-endian format.

2.1 Lidar Packet Header Definition

The lidar packet header occupies two 64 bit words.

Field Name	Fixed Point Representation	Units	Number of Bits	Start Bit	End Bit
Packet version - major	UQ8.0	Unitless	8	0	7
Packet version - minor	UQ8.0	Unitless	8	8	15
Packet version - patch	UQ8.0	Unitless	8	16	23
Packet sequence	UQ8.0	Incrementing value	8	24	31
Number of Rays in payload	UQ8.0	Count	8	32	39
Frame sequence	UQ8.0	Incrementing value	8	40	47
PTP Timestamp	UQ48.0	Seconds	48	48	95
Sensor ID	UQ8.0	Unitless	8	96	103
Data qualifier	UQ8.0	Unitless index, see description	8	104	111
Scan identifier	UQ16.0	Unitless	16	112	127

2.2 Ray Header

The ray header occupies two 64 bit words.

Field Name	Fixed Point Representation	Units	Number of Bits	Start Bit	End Bit
Azimuth angle	Q2.14	Radians	16	0	15
Elevation angle	Q2.14	Radians	16	16	31
PTP Timestamp	UQ32.0	Nanoseconds	32	32	63
Scan Checkpoint	User specified	User specified	8	64	71
Ray Sequence	UQ4.0	Incrementing value	4	72	75
Number of returns	UQ4.0	Count	4	76	79

Field Name	Fixed Point Representation	Units	Number of Bits	Start Bit	End Bit
Detector number	UQ1.0	Detector 0 or 1	1	80	80
Blockage level	UQ4.0	Level	4	81	84
RESERVED	Unspecified	-	2	85	86
Line number	UQ9.0	Incrementing value, resets to 0 at frame boundary	9	87	95
RESERVED	Unspecified	-	32	96	127

2.3 Return Data

The return data header occupies one 64 bit word.

Field Name	Fixed Point Representation	Units	Number of Bits	Start Bit	End Bit
Existence probability	[0, 255]	0 (0% probability) -> 255 (100% probability)	8	0	7
Range	UQ12.12	Meters	24	8	31
Reflectance	UQ1.15	Decimal percentage relative to a Lambertian target	16	32	47
Retro artifact	UQ1.0	Return flagged as potential retro artifact	1	48	48
Range wrap artifact	UQ1.0	Return flagged as potential range wrap artifact	1	49	49
RESERVED	Unspecified	-	1	50	50
Min range artifact	UQ1.0	Return flagged as potential min range artifact	1	51	51
Crosstalk artifact	UQ1.0	Return flagged as potential crosstalk artifact	1	52	52
Low Reflectance point	UQ1.0	Return flagged as ambiguous	1	53	53
RESERVED	unspecified	-	10	54	63

2.4 Data Protection Footer

The data protection footer occupies three 64 bit words at the end of the packet. This is a customer specific field containing checksums, CRCs, etc. for ensuring data integrity. The contents of this field are not detailed in this document.

2.5 Lidar Packet Header Field Descriptions

The lidar packet header describes the version of the data format, the number of rays in the packet, and other important information that applies to all the rays and returns in the packet.

2.5.1 Packet Version – major

The Iris data specification followed Semantic Versioning 2.0.0 [<https://semver.org/>]. A change to the MAJOR version marks an incompatible API change to the previous major version.

2.5.2 Packet version – minor

The Iris data specification followed Semantic Versioning 2.0.0 [<https://semver.org/>]. A change to the MINOR version adds functionality in a backwards compatible manner. For example, previously reserved bits may be defined to represent a new data type.

2.5.3 Packet version – patch

The Iris data specification followed Semantic Versioning 2.0.0 [<https://semver.org/>]. A change to the PATCH version fixes a bug in a backwards compatible manner.

2.5.4 Packet Sequence

This sequence number will increment once per Ethernet data packet, and will wrap back to 0 on the next packet after the maximum value is reached. Missing values in this counter indicate dropped Ethernet packets and lost data.

2.5.5 Number of Rays in payload

This value represents the number of rays in the payload.

2.5.6 Frame sequence

This sequence number will increment once per frame (as defined in the scan interface), and will wrap back to 0 on the next frame after the maximum value is reached.

2.5.7 PTP Timestamp - seconds

This value is the current system time in seconds, rounded down to the nearest integer. If PTP is active, this value will be synchronized with the PTP clock. If the sensor head has not been synchronized with PTP, the timestamp will be the time since the sensor head started, subject to local oscillator drift. If PTP synchronization is established and then lost, the timestamp will remain synchronized to the PTP master, subject to uncorrected local oscillator drift.

2.5.8 Sensor ID

This value is a customer configured field used to distinguish data between multiple sensors on a single vehicle. The default value is 0.

2.5.9 Data Qualifier

The data qualifier field expresses a self-assessment by the sensor of the validity of the lidar data in this packet. This field can change dynamically over time. Possible values are described in the table below.

Data Qualifier	Name	Meaning	Example Causes
0	DQ_Normal	Sensor is operating normally	Factory condition
1	DQ_Cat1Fault	Sensor has encountered a category 1 fault.	Optical scanner phase out of tolerance. See diagnostics reference for more details.

2.5.10 Scan Identifier

The scan identifier field is a reflection of the scan pattern's unique id followed at the time of ranging. The field is intended to ascribe lidar data to a unique or specific scan pattern. Support for this field is contingent on support in the SOME/IP command interface. See the associated service oriented architecture documentation for more details.

2.6 Ray Header Field Descriptions

The ray header describes information that is relevant for all returns in the ray, as well as the number of returns associated with the ray.

2.6.1 Azimuth Angle

The azimuth angle applies to all returns in this ray. Relative to looking "straight ahead" from the sensor's point of view, 0 is straight ahead, with angles increasing to the right.

2.6.2 Elevation Angle

The elevation angle applies to all returns in this ray. Relative to looking "straight ahead" from the sensor's point of view, 0 is straight ahead and level, with elevation angles increasing in the up direction.

2.6.3 PTP Timestamp - nanoseconds

This field is the nanosecond portion of the current system time stamp at the time that this ray was emitted from the sensor. If PTP is active, this value will be synchronized with the PTP clock. See the description of the PTP timestamp seconds field for a more detailed discussion.

The nanosecond portion of the timestamp is guaranteed to be monotonically non-decreasing per detector within a given packet. That is, for all returns belonging to a single detector (see field: [Detector number](#)), the timestamps will not decrease, but the data is not sorted by timestamp between detectors. For example, a packet may contain a ray from detector 0 with a nanosecond timestamp of 1000, followed by a ray from detector 1 with a nanosecond timestamp of 995.

This field resets to 0 at the beginning of a new second, which also triggers the creation of a new lidar data packet.

2.6.4 Scan Checkpoint

This field is supplied by the user in the scan control interface. It is returned without change.

2.6.5 Ray Sequence

This sequence number will increment once per ray, and will wrap back to 0 on the next ray after the maximum value is reached.

2.6.6 Number of Returns

This field indicates the number of returns that follow this ray header.

2.6.7 Detector Number

This field indicates which of the two detectors internal to the Iris sensor generated this ray and list of returns.

2.6.8 Blockage Level

This field indicates the level from 0 to 15 of window obscuration per ray. A level of 0 indicates no blockage, and a level of 15 indicates maximum blockage.

2.6.9 Line Number

This sequence number will increment once per scan line. It will wrap back to 0 on the next line after the maximum value is reached, and will reset to 0 if the frame sequence number changes.

2.7 Return Data Field Descriptions

Return data is repeated `Number of returns` times after the ray header, and describes the range and reflectivity of objects in the scene.

2.7.1 Existence Probability

Like any sensor, Iris is susceptible to certain noise and false alarms. These returns are largely detected through various filtering techniques, but are reported regardless. Returns with a high probability of being real world objects are reported with a high existence probability. Returns that are likely to be noise or false detections are reported with a low existence probability. It is suggested that users of the Iris sensor evaluate low probability returns against other sensors to determine if the data is real or a false alarm.

2.7.2 Range

This is the measured range to an object.

2.7.3 Reflectance

Simply put, this is an estimate of the reflectance of the object, measured as the ratio between the incident flux and the reflected flux at the reflecting surface. This estimate is independent of range but can vary with the incident angle at the surface. The sensor calculates the ratio between detected power and the power level expected for a 100% reflective Lambertian target at the detected range. Reflections from specular surfaces and retroreflectors will generally result in the ratio being greater than 100%.

If the beam hits near the edge of an object where not all of the beam energy hits the object, then consequently the reflected energy and thus the reported reflectance will be lower.

2.7.4 Retro artifact

Highly reflective surfaces can often return orders of magnitude more energy than typical materials, and therefore represent unique challenges. The retro artifact flag indicates returns which are suspected to either originate from, or be incorrectly located due to the presence of a retro reflector or specular reflection. There are other, more specific indicators for well defined artifacts that occur with retro reflectors documented below.

2.7.5 Range wrap artifact

Iris can have multiple pulses in flight, and as a result it is possible to detect energy returned from a previous pulse in certain circumstances. This is particularly prevalent on highly reflective overhead street signs. The range wrap flag indicates the return in question is suspected to originate from a preceding pulse. This means that it is likely a real return, but the target in question is at least 1 range gate further away than reported.

2.7.6 Min range artifact

Returns flagged as min range artifacts are suspected to be false alarms originating within the minimum range of the sensor.

2.7.7 Crosstalk artifact

Cross talk is an artifact which occurs at close range with highly reflective targets. It produces a well defined false alarm pattern. Points suspected to originate from this effect are tagged with the crosstalk flag.

2.7.8 Low Reflectance point

Certain very dim returns in the near field can be difficult to disambiguate from range wrap. These ambiguous points are tagged with the low reflectance flag. This tag does not necessarily indicate an artifact, simply that an issue may be present. It conveys that returns are from a very low reflectance object in the near field, but could also originate from a target at least one range gate further away than reported.

Revision Log

Rev. #	Date	Description	Initials
1.3.0	08/22/2023	▪ Added Scan Identifier	JK
1.2.2	08/04/2023	▪ Updated to match 1.2.2	JK / ZH
1.2.1	05/09/2023	▪ Updated to include new Return Data	JK / XW
1.1.0	08/01/2022	▪ Updated to include Blockage Level	JK
1.0.3	05/02/2022	▪ Updated to reflect current product	JK
1.0.1	06/24/2021	▪ Initial Release	JK